AUTOMATIC TABLET DISPENSER FOR COVID-19 PATIENT

COMPONENTS

Hardware compent:

1.Aruduino

2.Nodumcu(ESP8266)

3.Heart rate sensor

4.Buzzer

5.Bread board

6.LED display

7.Jumper wires

8.push button

9.RFID-RC522

10.Micro servo motor

SOFTWARE COMPONENTS:

1.Aruduino software

2.Circuit Digest cloud

Nodu MCU(ESP9266)WI FI module

```
#include <ESP8266WiFi.h>

#include <ESP8266HTTPClient.h>

#include <WiFiClientSecure.h>

#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
#include <SPI.h>
#include <MFRC522.h>

// WiFi credentials
const char* ssid = "hello";
const char* password = "00000000";

// API details
const char* apiKey = "uFJy0X5wRwij";
const char* templateID = "101";
const char* mobileNumber = "916374917133";
const char* var1 = "patient taking tablet";
const char* var2 = "collage";

// RFID and LCD setup
#define SS_PIN D2
#define RST_PIN D1
MFRC522 rfid(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
// Predefined user list
struct User {
    String uid;
    String name;
};

User users[] = {
    {"B6913A57", "Mathan"},
    {"83D46115", "vasan"},
    {"13D47F0D", "hari"}
};

String getUserName(String uid) {
    for (User user : users) {
        if (user.uid == uid) {
            return user.name;
        }
    }
    return "Unknown";
}
```

```
void setup() {
  Serial.begin(115200);
  Serial.println("Starting system...");
  WiFi.begin(ssid, password);

  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nConnected to WiFi");

  // Initialize SPI, RFID, and LCD
  Serial.println("Initializing SPI, RFID, and LCD...");
  SPI.begin();
  rfid.PCD_Init();
  Wire.begin(D3, D4);
  lcd.begin(16, 2);
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Scan Your Card");
```

```
  Serial.println("Ready to scan RFID cards.");
}

void sendSMS(String userName) {
  if (WiFi.status() == WL_CONNECTED) {
    WiFiClientSecure client;
    client.setInsecure();
    HTTPClient http;

    String apiUrl = "https://www.circuitdigest.cloud/send_sms?ID="
+ String(templateID);
    http.begin(client, apiUrl);
    http.addHeader("Authorization", apiKey);
    http.addHeader("Content-Type", "application/json");

    String payload = "{\"mobiles\":\"" + String(mobileNumber) +
"\",\"var1\":\"" + userName + " Access\",\"var2\":\"Granted\"}";
    Serial.println("Sending SMS: " + payload);

    int httpResponseCode = http.POST(payload);

    if (httpResponseCode == 200) {
```

```
        Serial.println("SMS sent successfully!");
      } else {
        Serial.print("Failed to send SMS. Error code: ");
        Serial.println(httpResponseCode);
      }
      http.end();
    } else {
      Serial.println("WiFi not connected! Cannot send SMS.");
    }
}

void loop() {
    if (rfid.PICC_IsNewCardPresent() && rfid.PICC_ReadCardSerial()) {
      Serial.print("Card detected! UID: ");
      String uid = "";
      for (byte i = 0; i < rfid.uid.size; i++) {
        uid += String(rfid.uid.uidByte[i], HEX);
      }
      Serial.println(uid);

      String userName = getUserName(uid);
```

```
        Serial.println("User Identified: " + userName);

        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Hello, " + userName);
        lcd.setCursor(0, 1);
        lcd.print("ID: " + uid);

        sendSMS(userName);

        delay(3000);
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Scan Your Card");
        Serial.println("Ready for next scan.");
    }
}
```

Arduino code

```
#include <SPI.h>
#include <MFRC522.h>
#include<LiquidCrystal_I2C.h>
```

```
#include<Wire.h>

#include <Servo.h>

int led = 13;

int ir = 3;

int ir2=4;

int ir3=5;

int ir4=6;

int voice=22;

int spk = 8;

#include<Servo.h>

int servopin1=11;

Servo myservo;

int servopin2=23;

Servo myservo2;

int pulsePin = A0;                 // Pulse Sensor purple wire connected to
analog pin A0

int blinkPin = 12;                // pin to blink led at each beat


// Volatile Variables, used in the interrupt service routine!

volatile int BPM;                   // int that holds raw Analog in 0. updated
every 2mS

volatile int Signal;                // holds the incoming raw data
```

```
volatile int IBI = 600;             // int that holds the time interval
between beats! Must be seeded!

volatile boolean Pulse = false;     // "True" when User's live heartbeat
is detected. "False" when not a "live beat".

volatile boolean QS = false;        // becomes true when Arduoino finds
a beat.


static boolean serialVisual = true;   // Set to 'false' by Default.  Re-set
to 'true' to see Arduino Serial Monitor ASCII Visual Pulse


volatile int rate[10];                    // array to hold last ten IBI values

volatile unsigned long sampleCounter = 0;         // used to determine
pulse timing

volatile unsigned long lastBeatTime = 0;          // used to find IBI

volatile int P = 512;                     // used to find peak in pulse wave,
seeded

volatile int T = 512;                     // used to find trough in pulse wave,
seeded

volatile int thresh = 525;                // used to find instant moment of
heart beat, seeded

volatile int amp = 100;                   // used to hold amplitude of pulse
waveform, seeded

volatile boolean firstBeat = true;        // used to seed rate array so we
startup with reasonable BPM
```

```
volatile boolean secondBeat = false;      // used to seed rate array so
we startup with reasonable BPM


 LiquidCrystal_I2C  lcd(0x27, 16, 2);

#define SS_PIN 9

#define RST_PIN 8

MFRC522 mfrc522(SS_PIN, RST_PIN);   // Create MFRC522 instance.

//Servo myServo; //define servo name


void setup()

{

  //Serial.begin(9600);   // Initiate a serial communication

  SPI.begin();      // Initiate  SPI bus

  mfrc522.PCD_Init();   // Initiate MFRC522

lcd.init();

  lcd.backlight();

  myservo.attach(servopin1);

  myservo2.attach(servopin2);


  myservo.write(85);

  myservo2.write(0); //servo start position
```

```
pinMode(blinkPin,OUTPUT);        // pin that will blink to your
heartbeat!
  Serial.begin(115200);            // we agree to talk fast!
  interruptSetup();


 pinMode(spk, OUTPUT);


  Serial.println("Put your card to the reader...");
  Serial.println();
  lcd.clear();
  lcd.setCursor(0,0);
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("1-Fever 2-cold");
    lcd.setCursor(0,1);
    lcd.print("3-cough 4-pain");
    delay(1000);
  pinMode(led,OUTPUT);
  pinMode(ir,INPUT);
  pinMode(ir2,INPUT);
  pinMode(ir3,INPUT);
  pinMode(ir4,INPUT);
```

```
  pinMode(voice,OUTPUT);


}
void loop()
{
  digitalWrite(spk,LOW);
 //myservo.write(50);
    serialOutput();
if(digitalRead(ir))
{
  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
 lcd.setCursor(0,0);
    lcd.print("  (1- FEVER)");
    lcd.setCursor(0,0);
    delay(3000);
    lcd.clear();
    lcd.print("hold-1 & 2-HIGH");
```

```
    lcd.setCursor(0,1);
    lcd.print("3-med  4-low");
    delay(1000);
    if(digitalRead(ir2))
{
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("    HIGH");
    //delay(3000);
    lcd.setCursor(0,1);

    lcd.print(".............");
    delay(3000);
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("please cheak the");
    //delay(3000);
```

```
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
```

```
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
      digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
```

```
delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
```

```
    myservo.write(50);
    delay(3000);
    myservo.write(85);



}

if(digitalRead(ir3))
{
 lcd.clear();
      digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);

    lcd.setCursor(0,0);
    lcd.print("med");
    //delay(3000);
    lcd.setCursor(0,1);

    lcd.print(".....");
```

```
delay(3000);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
```

```
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
```

```
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(50);
    delay(3000);
    myservo.write(85);

}
if(digitalRead(ir4))
{
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("low");
    //delay(3000);
```

```
lcd.setCursor(0,1);

lcd.print(".....");
delay(3000);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
```

```
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
```

```
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(50);
    delay(3000);
    myservo.write(85);

}
}

if(digitalRead(ir2))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
```

```
 lcd.clear();
lcd.setCursor(0,0);
   lcd.print("  (2-COLD)");
   lcd.setCursor(0,0);
   delay(3000);
   lcd.clear();
   lcd.print("hold-2 & 1-HIGH");
   lcd.setCursor(0,1);
   lcd.print("3-med  4-low");
   delay(1000);
   if(digitalRead(ir))
{
 lcd.clear();
   lcd.setCursor(0,0);
   lcd.print("    HIGH");
   //delay(3000);
   lcd.setCursor(0,1);

   lcd.print("............");
   delay(3000);
   lcd.clear();
```

```
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
```

```
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
```

```
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(115);
    delay(3000);
    myservo.write(85);



}

if(digitalRead(ir3))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("med");
    //delay(3000);
    lcd.setCursor(0,1);
```

```
    lcd.print(".....");
    delay(3000);
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("please cheak the");
    //delay(3000);
    lcd.setCursor(0,1);

    lcd.print("Heart rate");
    delay(5000);
serialOutputWhenBeatHappens();
    digitalWrite(led,HIGH);
    delay(100);
    digitalWrite(led,LOW);
    delay(100);
    digitalWrite(led,HIGH);
    delay(100);
    digitalWrite(led,LOW);
    delay(100);
    digitalWrite(led,HIGH);
```

```
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
```

```
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(115);
    delay(3000);
    myservo.write(85);

}
if(digitalRead(ir4))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("low");
```

```
//delay(3000);
lcd.setCursor(0,1);

lcd.print(".....");
delay(3000);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
```

```
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
```

```
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(115);
    delay(3000);
    myservo.write(85);

}
}
if(digitalRead(ir3))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
```

```
 lcd.clear();
lcd.setCursor(0,0);
   lcd.print("  (3-COUGH)");
   lcd.setCursor(0,0);
   delay(3000);
   lcd.clear();
   lcd.print("hold-3 & 1-HIGH");
   lcd.setCursor(0,1);
   lcd.print("2-med  4-low");
   delay(1000);
   if(digitalRead(ir))
{
 lcd.clear();
   lcd.setCursor(0,0);
   lcd.print("    HIGH");
   //delay(3000);
   lcd.setCursor(0,1);

   lcd.print("............");
   delay(3000);
   lcd.clear();
```

```
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
```

```
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
```

```
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(50);
    delay(3000);
    myservo.write(85);



}

if(digitalRead(ir2))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("med");
    //delay(3000);
    lcd.setCursor(0,1);
```

```
lcd.print(".....");
delay(3000);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart reat");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
```

```
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
```

```
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(50);
    delay(3000);
    myservo.write(85);

}
if(digitalRead(ir4))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("low");
```

```
//delay(3000);
lcd.setCursor(0,1);

lcd.print(".....");
delay(3000);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
```

```
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
```

```
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(50);
    delay(3000);
    myservo.write(85);

}
}
if(digitalRead(ir4))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
```

```
 lcd.clear();
lcd.setCursor(0,0);
   lcd.print("  (4-PAIN)");
   lcd.setCursor(0,0);
   delay(3000);
   lcd.clear();
   lcd.print("hold-4 & 1-HIGH");
   lcd.setCursor(0,1);
   lcd.print("2-med  3-low");
   delay(1000);
   if(digitalRead(ir))
{
 lcd.clear();
   lcd.setCursor(0,0);
   lcd.print("    HIGH");
   //delay(3000);
   lcd.setCursor(0,1);

   lcd.print("............");
   delay(3000);
   lcd.clear();
```

```
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart rate");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
```

```
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
  digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
```

```
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(180);
    delay(3000);
    myservo.write(85);



}

if(digitalRead(ir2))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("med");
    //delay(3000);
    lcd.setCursor(0,1);
```

```
lcd.print(".....");
delay(3000);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("please cheak the");
//delay(3000);
lcd.setCursor(0,1);

lcd.print("Heart reat");
delay(5000);
serialOutputWhenBeatHappens();
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
delay(100);
digitalWrite(led,LOW);
delay(100);
digitalWrite(led,HIGH);
```

```
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
digitalWrite(spk,LOW);
delay(100);
```

```
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(180);
    delay(3000);
    myservo.write(85);

}
if(digitalRead(ir3))
{  digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
 lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("low");
```

```
    //delay(3000);
    lcd.setCursor(0,1);

    lcd.print(".....");
    delay(3000);
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("please cheak the");
    //delay(3000);
    lcd.setCursor(0,1);

    lcd.print("Heart rate");
    delay(5000);
serialOutputWhenBeatHappens();
    digitalWrite(led,HIGH);
    delay(100);
    digitalWrite(led,LOW);
    delay(100);
    digitalWrite(led,HIGH);
    delay(100);
    digitalWrite(led,LOW);
```

```
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(led,HIGH);
  delay(100);
  digitalWrite(led,LOW);
  delay(100);
  digitalWrite(voice,LOW);
  delay(5000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("   Take the ");
  //delay(3000);
  lcd.setCursor(0,1);

  lcd.print("tablet");
  delay(1000);
    digitalWrite(spk,HIGH);
delay(100);
```

```
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    digitalWrite(spk,HIGH);
  delay(100);
  digitalWrite(spk,LOW);
  delay(100);
    myservo.write(180);
    delay(3000);
    myservo.write(85);

}
}


  // Look for new cards
  if ( ! mfrc522.PICC_IsNewCardPresent())
  {
```

```
  return;
}
// Select one of the cards
if ( ! mfrc522.PICC_ReadCardSerial())
{
  return;
}
//Show UID on serial monitor
Serial.print("UID tag :");
String content= "";
byte letter;
for (byte i = 0; i < mfrc522.uid.size; i++)
{
   Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
   Serial.print(mfrc522.uid.uidByte[i], HEX);
   content.concat(String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " "));
   content.concat(String(mfrc522.uid.uidByte[i], HEX));
}
Serial.println();
Serial.print("Message : ");
content.toUpperCase();
```

```
  if (content.substring(1) == "B6 91 3A 57") //change here the UID of
the card/cards that you want to give access
  {
    Serial.println("Authorized access");
    Serial.println();
    //delay(500);
    myservo2.write(90);
    delay(2000);
    myservo2.write(0);
    delay(1000);
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("Hi Mathan");
    lcd.setCursor(0,1);
    lcd.print("Raj");

    delay(3000);

   // delay(30*1000);

    lcd.clear();
    lcd.setCursor(0,0);
```

```
  lcd.print("1-Fever 2-cold");
  lcd.setCursor(0,1);
  lcd.print("3-cough 4-pain");
  delay(1000);





 // digitalWrite(LED_G, LOW);
}



  if (content.substring(1) == "83 55 44 1C") //change here the UID of
the card/cards that you want to give access
 {
  myservo2.write(90);
  delay(2000);
  myservo2.write(0);
  delay(1000);
  Serial.println("Authorized access");
  Serial.println();
  delay(500);
```

```
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Hi sethu");
  lcd.setCursor(0,1);
  lcd.print("Raman");

  delay(3000);

 // delay(30*1000);

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("1-Fever 2-cold");
  lcd.setCursor(0,1);
  lcd.print("3-cough 4-pain");
  delay(3000);

 }

else   {
  Serial.println(" Access denied");
```

```
  }



}



void interruptSetup()
{
  // Initializes Timer2 to throw an interrupt every 2mS.
  TCCR2A = 0x02;     // DISABLE PWM ON DIGITAL PINS 3 AND 11,
AND GO INTO CTC MODE
  TCCR2B = 0x06;     // DON'T FORCE COMPARE, 256 PRESCALER
  OCR2A = 0X7C;      // SET THE TOP OF THE COUNT TO 124 FOR
500Hz SAMPLE RATE
  TIMSK2 = 0x02;     // ENABLE INTERRUPT ON MATCH BETWEEN
TIMER2 AND OCR2A
  sei();             // MAKE SURE GLOBAL INTERRUPTS ARE ENABLED
}

void serialOutput()
{   // Decide How To Output Serial.
```

```
 if (serialVisual == true)
  {
    arduinoSerialMonitorVisual('-', Signal);   // goes to function that
makes Serial Monitor Visualizer
  }
 else
  {
     sendDataToSerial('S', Signal);     // goes to sendDataToSerial
function
   }
}


void serialOutputWhenBeatHappens()
{
 if (serialVisual == true) //  Code to Make the Serial Monitor
Visualizer Work
   {
    Serial.print(" Heart Beat Rate ");  //ASCII Art Madness
    Serial.print("BPM: ");
    Serial.println(BPM);
    lcd.clear();
    delay(3000);
```

```
    lcd.print("Heart Beat Rate ");
    lcd.setCursor(1,1);
    lcd.print("BPM: ");
    lcd.setCursor(5,1);
    lcd.print(BPM);
    delay(3000);
    lcd.clear();
  }
 else
  {
    sendDataToserial('B',BPM);   // send heart rate with a 'B' prefix
    sendDataToserial('Q',IBI);   // send time between beats with a 'Q' prefix
  }
}

void arduinoSerialMonitorVisual(char symbol, int data )
{
  const int sensorMin = 0;      // sensor minimum, discovered through experiment
  const int sensorMax = 1024;    // sensor maximum, discovered through experiment
```

```
  int sensorReading = data; // map the sensor range to a range of 12
options:
  int range = map(sensorReading, sensorMin, sensorMax, 0, 11);
  // do something different depending on the
  // range value:
}



void sendDataToSerial(char symbol, int data )
{
   Serial.print(symbol);
   Serial.println(data);
}

ISR(TIMER2_COMPA_vect) //triggered when Timer2 counts to 124
{
  cli();                                  // disable interrupts while we do this
  Signal = analogRead(pulsePin);             // read the Pulse Sensor
  sampleCounter += 2;                        // keep track of the time in mS
with this variable
  int N = sampleCounter - lastBeatTime;       // monitor the time since
the last beat to avoid noise
```

```
                                        //  find the peak and trough of the pulse
wave
  if(Signal < thresh && N > (IBI/5)*3) // avoid dichrotic noise by
waiting 3/5 of last IBI
    {
      if (Signal < T) // T is the trough
      {
        T = Signal; // keep track of lowest point in pulse wave
      }
    }

  if(Signal > thresh && Signal > P)
    {         // thresh condition helps avoid noise
      P = Signal;                          // P is the peak
    }                                     // keep track of highest point in pulse wave

  //  NOW IT'S TIME TO LOOK FOR THE HEART BEAT
  // signal surges up in value every time there is a pulse
  if (N > 250)
  {                                // avoid high frequency noise
    if ( (Signal > thresh) && (Pulse == false) && (N > (IBI/5)*3) )
      {
```

```
    Pulse = true;                          // set the Pulse flag when we think
there is a pulse
    digitalWrite(blinkPin,HIGH);               // turn on pin 13 LED
    IBI = sampleCounter - lastBeatTime;        // measure time
between beats in mS
    lastBeatTime = sampleCounter;              // keep track of time for
next pulse

    if(secondBeat)
    {                     // if this is the second beat, if secondBeat == TRUE
      secondBeat = false;                // clear secondBeat flag
      for(int i=0; i<=9; i++) // seed the running total to get a realisitic
BPM at startup
      {
        rate[i] = IBI;
      }
    }

    if(firstBeat) // if it's the first time we found a beat, if firstBeat ==
TRUE
    {
      firstBeat = false;                 // clear firstBeat flag
      secondBeat = true;                 // set the second beat flag
```

```
      sei();                          // enable interrupts again
      return;                         // IBI value is unreliable so discard it
    }
  // keep a running total of the last 10 IBI values
  word runningTotal = 0;              // clear the runningTotal variable

  for(int i=0; i<=8; i++)
    {             // shift data in the rate array
      rate[i] = rate[i+1];            // and drop the oldest IBI value
      runningTotal += rate[i];        // add up the 9 oldest IBI values
    }

  rate[9] = IBI;                      // add the latest IBI to the rate array
  runningTotal += rate[9];            // add the latest IBI to
runningTotal
  runningTotal /= 10;                 // average the last 10 IBI values
  BPM = 60000/runningTotal;           // how many beats can fit
into a minute? that's BPM!
  QS = true;                          // set Quantified Self flag
  // QS FLAG IS NOT CLEARED INSIDE THIS ISR
  }
}
```

```
  if (Signal < thresh && Pulse == true)
   {   // when the values are going down, the beat is over
     digitalWrite(blinkPin,LOW);           // turn off pin 13 LED
     Pulse = false;                        // reset the Pulse flag so we can do it
again
     amp = P - T;                          // get amplitude of the pulse wave
     thresh = amp/2 + T;                   // set thresh at 50% of the
amplitude
     P = thresh;                           // reset these for next time
     T = thresh;
   }

  if (N > 2500)
   {                          // if 2.5 seconds go by without a beat
     thresh = 512;                         // set thresh default
     P = 512;                              // set P default
     T = 512;                              // set T default
     lastBeatTime = sampleCounter;         // bring the lastBeatTime up
to date
     firstBeat = true;                     // set these to avoid noise
     secondBeat = false;                   // when we get the heartbeat back
```

```
    }

  sei();                                // enable interrupts when youre done!
}// end isr
```